

Task 8

Mini Project: Automated EC2 Backup System

As part of your cloud training, you are assigned a mini project called “**Automated EC2 Backup System.**”

This assignment is designed to help you understand how **Amazon EC2, AWS Lambda, Amazon Event Bridge, and IAM** work together to automate backups in a production-like environment.

Project Concept: Automated EC2 Backup System

In real-world cloud environments, backups must run **automatically, reliably, and without manual effort.**

In this project, you will design a system where **EC2 instances are backed up daily using EBS snapshots**, driven entirely by **serverless automation.**

Only EC2 instances that are **explicitly marked for backup** should be included in the process.

The backup operation must:

- Run on a fixed schedule
- Detect eligible EC2 instances
- Create snapshots automatically
- Record execution details in logs

Workflow

- Event Bridge triggers the Lambda function on schedule
- Lambda scans EC2 instances using tags
- Eligible instances are selected
- Attached EBS volumes are identified
- Snapshots are created automatically
- Execution details are logged
- The process repeats on the next scheduled run

Output Criteria

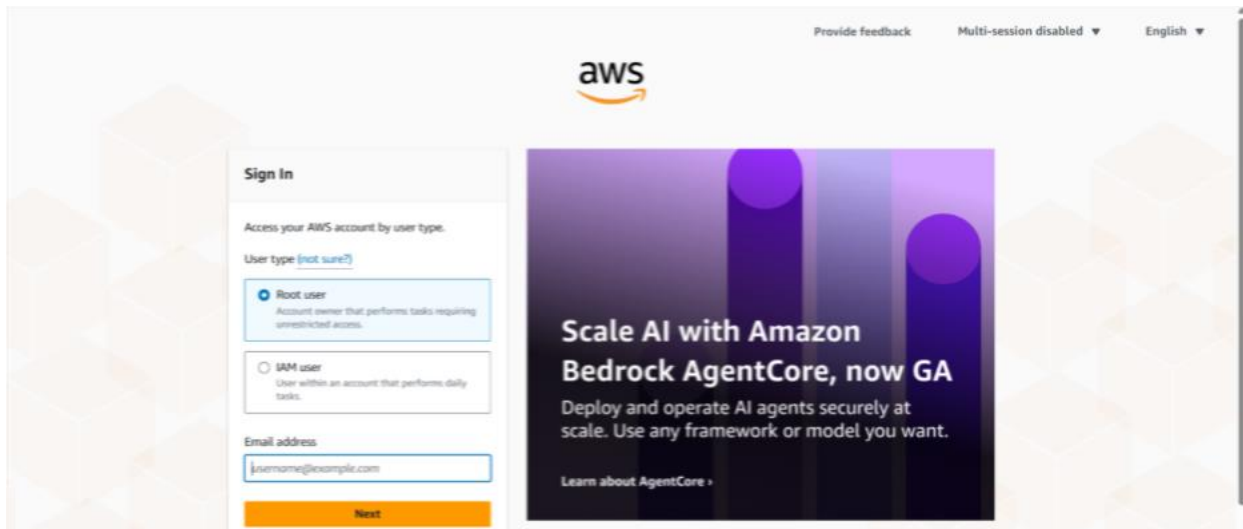
The task will be considered complete if:

- EC2 snapshots are created automatically
- Backup runs on a schedule without manual intervention
- Only tagged EC2 instances are backed up
- Lambda execution logs are visible
- IAM configuration follows best practices

Solution:

Login to your AWS account with credentials and required verification.

1. (<https://signin.aws.amazon.com/>)
My account – AWS Management Console- Sign in



2. Select Ec2 Instance to add tags.
 - Go to EC2 Console
 - Select the EC2 instance you want backed up
 - Click Tags → Manage tags
 - Add:
 - Key: Backup
 - Value: True
 - Save

EC2 > Instances > i-0cfc74bc93435c858 > Manage tags

Manage tags info
 A tag is a custom label that you assign to an AWS resource. You can use tags to help organize and identify your instances.

Key Value - optional [Remove](#)

[Add new tag](#)
 You can add up to 49 more tags.

[Cancel](#) [Save](#)

3. Create a IAM role for Lambda Function.

- Go to IAM → Roles → Create role
- Select Trusted entity: AWS service
- Choose Use case: Lambda
- Click Next
- Attach Permission
 - *AmazonEC2ReadOnlyAccess*
 - *AmazonEC2FullAccess (needed for snapshots)*
 - *CloudWatchLogsFullAccess*
- Create Role

IAM > Roles > Create role

Step 2
 ○ Add permissions
 Step 3
 ○ Name, review, and create

Trusted entity type

☒ **AWS service**
 Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
 Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
 Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
 Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
 Create a custom trust policy to enable others to perform actions in this account.

Use case
 Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Choose a use case for the specified service.
Use case

☒ **Lambda**
 Allows Lambda functions to call AWS services on your behalf.

☐ **AWSServiceRoleForLambda**
 Allows Lambda to manage AWS resources on your behalf.

IAM > Roles > Create role

```

10
11
12
13
14
15
16
    }
    "lambda.amazonaws.com"
  }
}

```

Step 2: Add permissions [Edit](#)

Permissions policy summary

Policy name	Type	Attached as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonEC2ReadOnlyAccess	AWS managed	Permissions policy
CloudWatchLogsFullAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)
 Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
 No tags associated with the resource.

[Add new tag](#)

4. Create lambda Function

- Go to AWS Lambda → Create function
- Choose Author from scratch
- Function name: *EC2_Automated_Backup*
- Runtime: *Python 3.14*
- Execution role:
 - Select Use existing role
 - Choose LambdaEC2BackupRole
- Click Create function

Lambda > Functions > Create function

Common use cases:

Basic information

Function name
 Enter a name that describes the purpose of your function.

 Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
 Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
 [Last fetched 12/31/2025, 3:19:45 PM](#)

Durable execution - new [Info](#)
 Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).
☐ Enable

Architecture [Info](#)
 Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

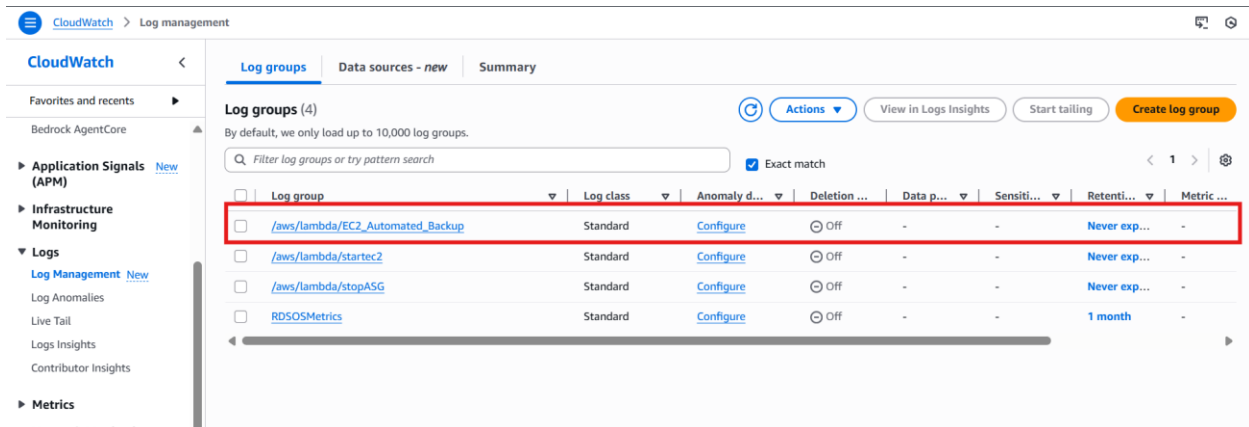
Permissions [Info](#)
 By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

Execute the code in lambada function



5. Verify CloudWatch Logging

- Go to CloudWatch → Logs → Log groups
- Find: `/aws/lambda/EC2_Automated_Backup`
- Logs will appear after Lambda runs



6. Create EventBridge Rule (for Automation)

- Go to Amazon EventBridge - Rules
- Click Create rule
- Rule name: `DailyEC2BackupRule`
- Rule type: `Schedule`
- Choose Cron expression
`cron(30 11 31 12 ? 2025)/` Change as per your need
 Used this cron job for Today at 5 PM IST (11:30 UTC)

- Select Targets

AWS Services – Lambda Function - EC2_Automated_Backup

The screenshot shows the Amazon EventBridge console. On the left is a navigation menu with links to Dashboard, Developer resources (Learn, Sandbox, Quick starts), Buses (Event buses, Rules, Global endpoints, Archives, Replays), Pipes, Scheduler (Schedules, Schedule groups), and Integration. The main content area has a dark header with 'Amazon EventBridge' and the tagline 'A serverless service for building event-driven applications'. Below this is a 'Get started' section with four options: EventBridge Rule with event pattern, EventBridge Scheduled rule (selected), EventBridge Pipe, and EventBridge Schema registry. A 'Create scheduled rule' button is at the bottom right. A 'Pricing' link is also visible.

This screenshot shows the 'Define rule detail' step of the 'Create rule' wizard. The left sidebar shows the progress: Step 1 (Define rule detail, active), Step 2 (Define schedule), Step 3 (Select target(s)), Step 4 (optional: Configure tags), and Step 5 (Review and create). The main form includes: a 'Name' field with 'DailyEC2BackupRule'; a 'Description - optional' field with 'daily backup of ec2'; an 'Event bus' dropdown set to 'default'; and a checked option 'Enable the rule on the selected event bus'. A blue callout box promotes 'EventBridge Scheduler - A new AWS scheduling capability!'. At the bottom are 'Cancel' and 'Next' buttons.

This screenshot shows the 'Define schedule' step of the 'Create rule' wizard. The left sidebar shows the progress: Step 1 (Define rule detail), Step 2 (Define schedule, active), Step 3 (Select target(s)), Step 4 (optional: Configure tags), and Step 5 (Review and create). The main form includes: a 'Schedule pattern' section with two radio buttons, the first of which is selected; a 'Cron expression' section with a cron field set to '30 11 31 12 ? 2025' and a 'Next 10 trigger date(s)' field showing 'Wed, Dec 31, 2025, 05:00 PM GMT+5:30'. At the bottom are 'Cancel', 'Previous', and 'Next' buttons.

Amazon EventBridge > Rules > Create rule

Step 5
Review and create

Amazon EventBridge

- Dashboard
- Developer resources
 - Learn
 - Sandbox
 - Quick starts
- Buses
 - Event buses
 - Rules **Updated**
 - Global endpoints
 - Archives
 - Replays
- Pipes
 - Pipes
- Scheduler
 - Schedules
 - Schedule groups

Target type

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus
☐ EventBridge API destination
☒ AWS service

Select a target [Info](#)

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function

Target location

☒ Target in this account
☐ Target in another AWS account

Function

EC2_Automated_Backup

Configure version/alias

Permissions

☒ Use execution role (recommended)

Execution role

EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. [EventBridge and AWS Identity and Access Management](#)

☒ Create a new role for this specific resource
☐ Use existing role

Role name

First test manually if this working

Snapshots (2) [Info](#)

Last updated 8 minutes ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

Owned by me

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Starte
☐		snap-0e6f289683e13d9a6	-	8 GiB	Automated backup of i-01...	Standard	Pending	2025/
☐		snap-00c05319d120ac0b1	1.62 GiB	8 GiB	Automated backup of i-01...	Standard	Completed	2025/

Log streams (4) [Delete](#) [Create log stream](#) [Search all log streams](#)

By default, we only load the most recent log streams.

☐ Exact match ☐ Show expired [Info](#)

☐	Log stream	Last event time
☐	2025/12/31/[\$LATEST]b411fe333468487cbbb18be3f64c62a7	2025-12-31 11:10:09 (UTC)
☐	2025/12/31/[\$LATEST]476b15020a8b48ea84017d2f6423a599	2025-12-31 10:49:15 (UTC)
☐	2025/12/31/[\$LATEST]83d806c669574cf7a7eca41beb55f2c1	2025-12-31 10:27:20 (UTC)
☐	2025/12/31/[\$LATEST]91c0213d6efa4a1daa425b7abd78afb0	2025-12-31 10:08:33 (UTC)

After Setting Cron Job for specific time (05:05 PM)

Snapshots (3) [Info](#)

Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

Owned by me

☐	Snapshot ID	Full sn...	Volu...	Description	Storage tier	Snapshot status	Started	Progress
☐	snap-0818987231007e915	-	8 GiB	Automated backup of i-01...	Standard	Pending	2025/12/31 17:05 GMT+5:30	0%
☐	snap-0e6f289683e13d9a6	1.62 GiB	8 GiB	Automated backup of i-01...	Standard	Completed	2025/12/31 16:56 GMT+5:30	100%
☐	snap-00c05319d120ac0b1	1.62 GiB	8 GiB	Automated backup of i-01...	Standard	Completed	2025/12/31 16:40 GMT+5:30	100%

Log streams (6)

By default, we only load the most recent log streams.



Delete

Create log stream

Search all log streams

Q Filter log streams or try prefix search

☐ Exact match

☐ Show expired [Info](#)

< 1 > ⚙

☐ Log stream	Last event time
☐ 2025/12/31/[\$LATEST]5de307307af74e59836fc4b2218546d5	2025-12-31 11:35:07 (UTC)
☐ 2025/12/31/[\$LATEST]eecd5b9be7ec416cb9ad81378dcb3002	2025-12-31 11:26:27 (UTC)
☐ 2025/12/31/[\$LATEST]b411fe333468487cbbb18be3f64c62a7	2025-12-31 11:10:09 (UTC)
☐ 2025/12/31/[\$LATEST]476b15020a8b48ea84017d2f6423a599	2025-12-31 10:49:15 (UTC)
☐ 2025/12/31/[\$LATEST]83d806c669574cf7a7eca41beb55f2c1	2025-12-31 10:27:20 (UTC)
☐ 2025/12/31/[\$LATEST]91c0213d6efa4a1daa425b7abd78afb0	2025-12-31 10:08:33 (UTC)